
VARIATIONS IN SOCIOECONOMIC DEVELOPMENT BY REGION

Interregional Inequality in Russia and Post-Soviet Countries in the 21st Century¹

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Abstract—The article analyzes the level and dynamics of interregional inequality in Russia, Kazakhstan, Uzbekistan, and Belarus versus other European countries based on data from official statistics of CIS countries (1995–2022) and Eurostat (2011–2022). The Gini coefficient, weighted by population and normalized depending on the number of territorial units, is used as the main assessment parameter. In Belarus and European Union countries, the level of interregional inequality is lower, associated with high and more uniform development of the territory and a more stable economic structure. In Russia, Kazakhstan and Uzbekistan, the level of interregional inequality is higher and its dynamics are multidirectional, associated with the influence of resource rent on a limited number of regions. In Russia and Uzbekistan since the late 2010s, inequality has grown, decreased in Kazakhstan, and was small and stable in Belarus due to the peculiarities of the economic structure and institutional factors of development. Assessments of the impact of macroeconomic dynamics on interregional inequality have not provided clear evidence of its impact. The impact of the state's redistribution policy can only be assessed for Russia: a significant increase in transfers during the crises of 2009 and 2020 contributed to mitigating inequality. Within Russian federal districts, differentiation is generally lower. High inequality is characteristic of the extremely heterogeneous Ural Federal District and Central Federal District, where it has been declining in recent years. In the Far Eastern Federal District, regional differentiation grew, and its gradual increase also occurred in the regions of the Northwestern Federal District.

Keywords: interregional inequality, GRP, investments, personal money income, poverty level, Russia, Kazakhstan, Uzbekistan, Belarus

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INTRODUCTION AND FORMULATION OF THE PROBLEM

The problem of territorial inequality attracts the attention of scientists, politicians, and society, since it has both an economic and sociopolitical perspective. Large gaps in the level of development of territories are perceived negatively; in the public sphere, there is much speculation and emotional judgment on this topic. Science has long proven that territorial inequality is inevitable; it depends on the competitive advantages and barriers to the development of certain territories (Krugman, 1991). The question is its scale and

dynamics. This is why correct scientific assessments of territorial inequality are so important.

Research into interregional inequality in Russia is carried out mainly by economists and much less frequently by geographers. There are many publications, and most of them state the growth of inequality, although sometimes it is not inequality itself that is considered, but the processes of territorial concentration, which is not the same thing. This is shown by the literature review in the next section of the article.

Previous publications by the authors of this article show that the dynamics of interregional inequality is

¹ The journal *Regional Research of Russia* traditionally publishes English translations of Russian-language articles published in the journal *Regiona'nye Issledovaniya*. This article is one of four similar examples in this issue, and the editorial board of *Regional Research of Russia* considers it useful to present its readers with *Regiona'nye Issledovaniya*. The journal was founded in 2002 by the Institute of Geography of the Russian Academy of Sciences; Immanuel Kant Baltic Federal University, and Smolensk State University and is published quarterly; About 1000 scientific articles have already been published, devoted to methodology, theory, methods, and practice of regional research in Russia and abroad. This is the only journal in Russia focused primarily on economic, social, cultural, political, and recreational geography. Articles on regional and spatial economics, regional sociology, regional analysis, and regional policy also occupy an important place. In the Russian SCIENCE INDEX ranking, the journal ranks (2022) seventh among journals on the subject of "Geography"; according to the two-year impact factor by the Russian Index of Science Citation (RINTs) is the most cited among Russian journals on the subject of "Geography"; it ranks third and fourth on the h-index over the past 10 years.

not linear and changes under the influence of various factors, including the macroeconomic background and redistribution policy of the state (Zubarevich and Safronov, 2013). These findings were based on an analysis of data from the second half of the 1990s to the early 2010s. More than a decade has passed since then, during which a series of crises occurred in Russia. It is important to understand whether economic turbulence has affected interregional inequality and, if so, how? At the same time, it is better to measure inequality using a wide range of economic and social indicators in order to assess differences in its level and dynamics in different areas of development. In addition, comparative studies of interregional inequality in various post-Soviet countries, especially larger ones, versus European Union countries, are increasingly relevant in understanding Russia's characteristics a broader context. Another noteworthy perspective is economic inequality between regions within individual federal districts (FDs) or economic areas, which is often assessed by comparing individual statistical indicators without using standard methods for specifically measuring inequality. This article attempts to consider all these areas of assessing interregional inequality to compare trends over the long time period since the 1990s up to 2022.

REVIEW OF EARLIER STUDIES

Interregional inequality in Russia and similar topics have been studied to a greater extent in the period after the publication of (Zubarevich and Safronov, 2013); the review focuses on recent publications. They can be grouped by several dimensions, reflecting different aspects of measuring inequality. There is also an increasing number measurement methods, but their adequacy may vary. For these reasons, the review offers greater detail than is commonly found in articles.

The first direction measures the level of inequality and its dynamics. The first such study was conducted at the Gaidar Institute using data from the second half of the 1990s and early 2000s, with coefficients of variation to assess inequality (Drobyshevskii et al., 2005). It did not reveal distinct trends of growth or decline, due to the multidirectionality of the economic dynamics during this period. Research by Novosibirsk economists used the decomposition method to identify the contribution of individual sectors of the economy to regional inequality in per capita gross regional product (GRP) (Lavrovskii and Shil'tsin, 2009). In earlier works of the Novosibirsk economic school, the methodology for measuring the differentiation of economic development of regions using the GRP indicator was analyzed (Granberg et al., 1998). The methodology for measuring interregional differentiation of GRP in BRICS countries, which includes Russia, is also considered, taking into account the characteristics of national accounts in each of them and the quality of

statistics; it also shows a high level of territorial differentiation in these countries (Zaitseva, 2010).

In Russian geographical science, measurements of territorial inequality based on data for the mid-2000s were started by A.I. Treivish. His calculations of the unevenness of the population of four countries across 10–12 large regions showed that Russia is similar to China and the United States, and among large countries, only Canada stands somewhat apart (Treivish, 2009). The standard deviations of per capita GRP of the regions were also calculated as a percentage of national values. In the USA (51 regions), they were 38.4; in the European Union, depending on the number of regions, 42–48; in Mexico (32 regions), 53; in Brazil (27 regions), 57; in India (35 regions), 66; in Russia (79 regions without autonomous okrugs), 67; and in China, 71 (Treivish, 2009). Thus, less developed large countries are more contrasting internally, and Russia is among the leaders in interregional inequality.

A comparative analysis of regional inequality for a number of economic and social indicators was carried out for three large post-Soviet countries over a longer period using the Gini coefficient: from the mid-1990s to the end of the 2000s (Zubarevich and Safronov, 2011). The study showed that economic inequality in Russia grew until the mid-2000s, then began to decline, and according to income indicators it began to decline from 2003. In Kazakhstan, mitigation of inequality began later and not for all indicators; Ukrainian statistics showed hardly any effect. In a subsequent work by the authors, inequality was examined according to various indicators not only at the regional level, but also at the level of Russian cities for the period before 2011. It was shown that interregional inequality from the mid-2000s to the early 2010s continued to lessen, and intercity inequality, as measured by wages, showed stability and decreased slightly only when Moscow and St. Petersburg were excluded from the calculations (Zubarevich and Safronov, 2013).

Another study drew a conclusion about the long-term dynamics of increasing regional inequality (Shatalova and Kasatkina, 2022). The authors assessed inequality for various economic indicators using the integral decile coefficient. According to their calculations, for the period 1995–2020, the level of regional economic inequality increased from 2.4 to 5.6 times. However, Theil index plots presented in the same article show different inequality dynamics (both growth and decline) in GRP and investment for different periods (Shatalova and Kasatkina, 2022).

The inequality of federal districts and regions within them was also considered. For a number of socioeconomic development indicators for 2000–2019, calculations were carried out using the Theil index. The study found that differences between federal districts explain a small part of interregional differentiation, and differences between regions within districts account for more than 80% of all regional variation (Gagarina and Bolotov, 2021).

There are few studies of regional inequality in Kazakhstan, including a report by the United Nations Development Program based on an analysis of statistical indicators of regions (Regional ..., 2016). A study was also conducted of inequality at the level of municipalities, divided into groups according to the center–periphery model (degree of distance from regional centers). Calculations of variation coefficients for per capita indicators of industrial production, investment, housing construction, wages for the period from 2002 to 2017 showed that the differences between different groups of municipalities in industrial production were increasing, while in housing construction and retail trade, they were slowly decreasing; i.e., more peripheral municipalities are moving slightly closer to metropolitan municipalities (Iskaliyev, 2022).

The second direction comprises studies that do not evaluate interregional inequality itself, but are close to this topic in approaches and methods. These are mainly studies by the Novosibirsk economic school on territorial concentration and unevenness of economic activity. Recent studies examine the methodology for measuring regional indicators, the inequality of eastern regions, and necessary changes in regional policy (Suspitsyn, 2014, 2022).

For this article, research by E. Kolomak is more relevant, who calculated the Theil index for territorial population concentration, the number of employed people, and GRP for 1995–2011 in two territorial sections: for the West and East of the country, as well as at the regional level (Kolomak, 2014). The differences in GRP were greatest, although they decreased somewhat between West and East. At the same time, in West, differences between regions grew faster than in the East, both in population, employment levels, and GRP; i.e. spatial concentration of economic activity is more typical of Western Russia (Kolomak, 2014). In another study by this author, fixed assets were added to the economic activity indicators; calculations were carried out using the Theil index and multiple regression (Kolomak, 2019). The highest absolute values of the index were obtained for fixed assets and GRP; higher rates of growth of differences were also observed for fixed assets. The spatial concentration of labor resources, as well as the increase in differentiation in employment levels, was much lower. The work examined several directions of transformation of the territorial structure of these indicators: West–East, center–periphery, regions with mining and manufacturing specialization. It is shown that the overall concentration of economic activity is accompanied by migration of resources from the periphery to the center, from East to West, and from regions of mining to regions of manufacturing specialization, but the contribution of concentration to the overall level of interregional differences has remained small. The author concludes that in the economic development of Russia in the 21st century, spatial changes occur, but their relatively low rate speaks more of an evolutionary process.

Let us also mention a geographical study of the territorial concentration of GRP in European Union countries and Russia based on data for 2007–2015. It was aimed at identifying the role of world cities in this process using the decomposition of the Theil index (Antonov, 2020). The study revealed similar trends in population concentration and GRP, primarily due to global cities, as well as the significant role of fragmentation of the used grid of administrative-territorial units (ATU) in assessing the direction and intensity of these processes. For the regions of Russia, excluding the two capitals, multidirectional and unstable trends in the concentration/deconcentration of the population and economy were noted, and waves of economic crises had a noticeable impact on them.

There are publications with more clear conclusions. For example, there has been a steady increase in inequality not only in terms of economic development, but also in living standards (Kryukov and Kolomak, 2021). In support, data are provided on the growth rates of nominal and real GRP in federal districts from 1998 to 2017, changes in the shares of federal districts in the total volume of GRP, industrial production, number of employees, energy consumption (Kryukov and Kolomak, 2021). The statement about the steady growth of interregional differences requires discussion: over what period, at what speed, and how sustainable is this trend? It is these issues that will be discussed below when analyzing the calculations made by the authors of this article.

The polarization of Russian regions over two decades of the 21st century based on the dynamics of GRP and population was also assessed using more complex methods—using Markov processes with discrete time and continuous state space (Bufetova, 2022). At the same time, the author excluded from consideration not only the Chechen Republic, the Republic of Crimea and the city of Sevastopol² due to insufficient data, which is understandable, but also Moscow, Tyumen oblast with autonomous okrugs, St. Petersburg and Moscow oblast, which do not look convincing. Using complex econometrics, three groups of regions were identified: lagging, average performers, and leaders, with division into subgroups (e.g., one subgroup, quite strange for regionalists, consisted of only three regions with high GRP growth rates: the Republic of Dagestan and Sakhalin and Belgorod oblasts). The study draws conclusions about the increasing gap between groups, the increasing concentration of the population in the leading regions, the impossibility of regions moving from the category of lagging to the average performers, or from average performers to leaders, which indicates the presence of traps of low and average levels of development. Another conclusion that remains unspecified is that

² In the article, Russia's borders are considered in accordance with the Constitution of the Russian Federation adopted by popular vote on December 12, 1993, with amendments approved during the All-Russian vote on July 1, 2020.

during the period under study, polarization occurs within the largest group of average performers, which includes almost three-quarters of regions.

The third direction measures various aspects of interregional inequality, most often by income. Studies of this type also widely use econometric approaches, in particular, to analyze the contribution of various sources to regional income inequality (Malkina, 2017), as well as intraregional income inequality and its impact on economic growth in Russian regions for the period from 2005 to 2020 (Tumanyants et al., 2023). Using various indicators of intraregional income inequality and complex econometrics, the authors concluded that high intraregional income inequality corresponds to lower rates of economic growth in the region, but this dependence manifests itself only in the long term and is not visible over medium and short time intervals.

Measuring regional inequality in the tax and budget sphere using the Gini coefficient did not reveal clear overall trends in the dynamics for 2008–2022 (Zubarevich and Safronov, 2023). Calculations of inequality in tax revenues from the territories of federal subjects for 2016–2022, which correlated with the number of employees, were carried out using different methods (Gini, Theil, Atkinson, etc., indices) (Kamaletdinov and Ksenofontov, 2023). They also revealed nonlinear dynamics of inequality across all indicators and measurement methods.

A review of publications shows that, despite the expanded range of analytical methods used, the conclusions by the authors of different cited works often contradict each other in assessing interregional inequality dynamics.

MATERIALS AND METHODS

The information base consists of two blocks. The first includes available materials from state statistical bodies of large post-Soviet countries: Russia, Kazakhstan, Uzbekistan, and Belarus, which, along with data on GRP dynamics, cover a wider range of socioeconomic characteristics of regions, mainly for 1995–2022. In addition, data from the Federal Treasury of the Russian Federation were used to analyze individual indicators of the Russian budget. The second block comprises Eurostat data on GRP in the context of territorial units of the NUTS-2 level, most comparable in scale to federal subjects, for some European countries, and Turkey for 2011–2022, as well as World Bank data on the dynamics of the index of physical volume of their GDP.

An important point when considering territorial differentiation/inequality is the idea of the object of study. Some economists propose an approach to regions as separate points of equal weight (Glushchenko, 2016), but it is inadequate to the problem we formulated. Regions are considered not as abstract elements of an econometric model, but as places of life of the population in specific socioeconomic condi-

tions with different availability of basic services and benefits. Therefore, based on the experience of other studies, the Gini coefficient, weighted by the population of territorial units, was chosen as the main characteristic of the level of interregional differentiation as the most clearly interpretable indicator, which takes into account in a balanced manner the characteristics of the distribution of regions according to the analyzed indicators.

When analyzing one country, the most interesting thing is not so much the value of the coefficient as its dynamics. If it is necessary to compare territorial systems of different composition, it is necessary to take into account the difference in the number of regions, which, although not decisive, has a noticeable impact on the calculation results, especially for countries with a small number of ATUs; this is justly pointed out in (Glushchenko, 2015). Therefore, for intercountry comparisons, the Gini coefficients were normalized to a value corresponding to hypothetical absolute inequality taking into account the number of regions (n) in each country $[(n - 1)/n]$.

RESULTS

(1) To understand the level and dynamics of interregional inequality in Russia, a broader context is needed. Without pretending to provide a deep analysis of the topic of interregional inequality in European countries, on which there are many publications, for the purpose of subsequent comparison with Russia, we will carry out calculations of inequality in European countries and its changes since the 2010s, according to GRP at the regional level NUTS-2. Calculations show that in Turkey, in the less developed Balkan countries and Hungary it is more significant, and not only the most developed countries have minimum values, but also the small and less developed Portugal³ (Fig. 1). In almost all of these countries, inequality has been stable or has fallen slowly since the 2010s. Thus, in more developed European countries, regional economic inequality is lower and stable or declining.

Methodological studies examining the use of the Gini coefficient to assess regional inequality indicate that this indicator is influenced by the different number of territorial units in countries (Glushchenko, 2015). Normalization taking into account the number of regions of the Gini coefficient leads to its not very significant increase for countries with a small number of territorial units. Hungary becomes the leader in terms of interregional economic inequality, while Bulgaria and Serbia are slightly ahead of Turkey. This does not change the main substantive conclusion: in gen-

³ A.I. Treivish explained to the authors of the article that Portugal is contrasted along the “mountainous east–coastal west” axis, while NUTS-2 divides the country from south to north, so the NUTS-3 level (more fractional) is more adequate for measuring differences.

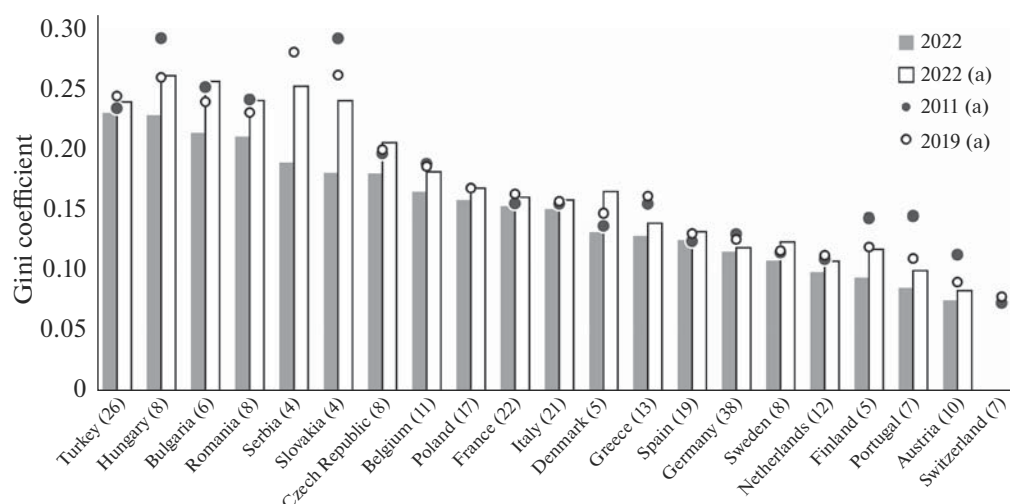


Fig. 1. Gini coefficient for interregional GRP inequality 2011–2022.

Note: number of territorial units for NUTS-2 level is indicated in parentheses; (a) normalized.

Source: authors' calculations based on Eurostat data.

eral, in European countries, even in relatively less developed ones, the level of interregional inequality is low.

Also, for comparison with Russia, we calculated the annual values of the Gini coefficient for interregional economic inequality in GRP for large post-Soviet and two Eastern European states (Poland and Romania), which show that its level is much higher in Russia and Kazakhstan (Fig. 2). This is a consequence not only of the size of the countries, but also of strong territorial contrasts in the availability of mineral resources and the development of space. However, still, since the mid-2000s inequality in them began to decrease. In Kazakhstan, it has decreased even more significantly since the early 2020s, and as a result, the two countries have become equal in terms of interregional GRP inequality. The dynamics of recent years in Kazakhstan are associated with changes in regional policy in the late 2010s. Uzbekistan has moved closer to Russia and Kazakhstan in recent years, where inequality increased only during periods of accelerated economic growth in the 2000s, characteristic of all post-Soviet countries, then from the late 2010s, due to changes in economic policy and the subsequent behind this economic growth. In Romania, Belarus and Poland, inequality is lower and stable, which is closer to the situation in other European countries. Stability is facilitated by both higher population density and development of the territory, as well as the absence of raw material rent for the vast majority of regions, which is subject to sharp price fluctuations.

(2) The influence of the *dynamics of economic development of countries* on interregional inequality is not very large and is ambiguous even for two resource-producing countries: Russia and Kazakhstan, in which interregional differentiation is largely related to natural resource rent and the situation in world prices for hydrocarbon raw materials. It cannot be said that

rapid economic growth leads to increased interregional inequality, and vice versa (Fig. 3a). Therefore, in the crisis of the 1990s, interregional inequality grew despite the negative economic dynamics, since regions without exportable natural resources fell more strongly. In the 2000s higher growth rates of the index of physical volume of GDP in Russia and Kazakhstan were largely ensured by a favorable environment for raw material export regions, which was accompanied by an increase in interregional differentiation, but only until the second half of the 2000s. In the 2010s, despite different economic dynamics, interregional inequality was relatively stable and even decreased slightly. Turbulent economic dynamics in the early 2020s led to different results: in Kazakhstan, interregional inequality decreased slightly, while in Russia it increased.

In Uzbekistan, economic growth in the 2000s was accompanied by increasing interregional inequality until the 2009 crisis, and in the 2010s, with a slight slowdown in economic growth, inequality decreased. Subsequent increases in interregional inequality since the late 2010s occurred under different economic dynamics, there is an inconsistency in trends.

The relationship with economic dynamics is even less noticeable for countries with lower levels of inequality (Fig. 3b). In Poland, the range of fluctuations in economic dynamics is significantly lower than in resource-producing countries, and has virtually no effect on the differentiation of regions in the conditions of an old-developed and densely populated space. In Romania and Belarus, the economies are less developed and less stable, but the situation is similar—there is no clear connection between the dynamics of gross domestic product (GDP) and interregional economic inequality (Gini coefficient for GRP).

Thus, we cannot claim that interregional inequality is clearly influenced by the macroeconomic factor (GDP dynamics). Rather, this is the result of the

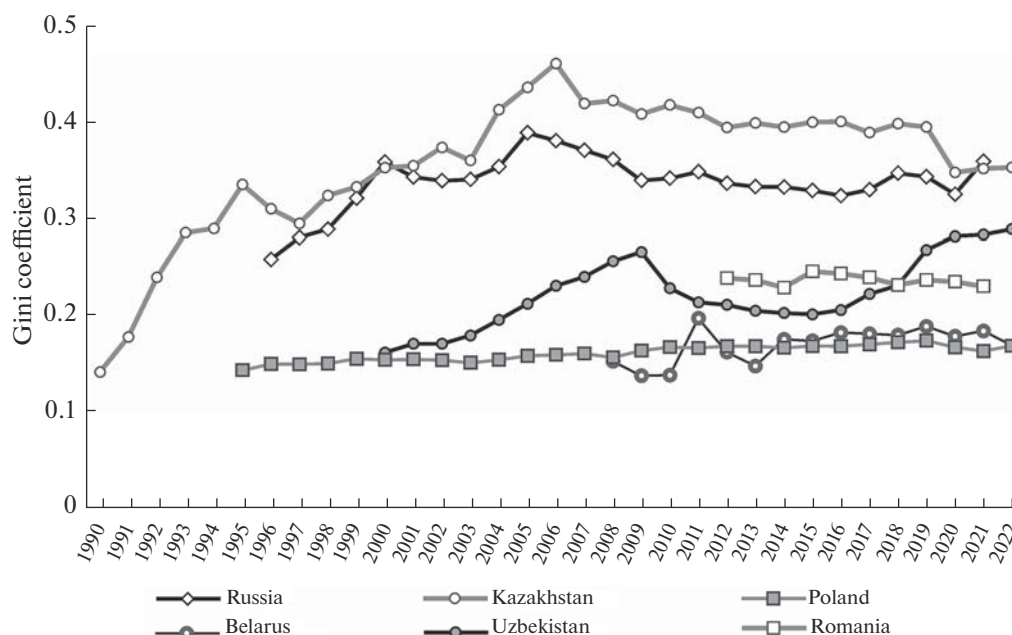


Fig. 2. Normalized Gini coefficient for economic inequality (GRP) for some post-Soviet countries and Central and Eastern European Countries.

Source: authors' calculations based on data from Rosstat, Bureau of National Statistics of Republic of Kazakhstan, National Statistical Committee of Republic of Belarus, Statistics Agency of the Republic of Uzbekistan, and Eurostat.

cumulative the actions of a wider range of factors, both exogenous and endogenous, including those related to the structural features of the regional economy.

(3) A significant factor influencing interregional inequality is the *redistribution policy of the state*. Large-scale support for less developed regions in the form of transfers smooths out interregional economic inequality, since regional budget expenditures for social purposes are taken into account when measuring GRP as part of household final consumption expenditures. The impact of this factor can only be assessed for Russian regions, for which budget statistics are available. The dynamics of transfers were different: they grew fastest during the years of implementation of national projects (2007) and during the crises of 2009 and 2020. As previously conducted studies have shown, owing to transfers, regional inequality in per capita income and budget expenditures has softened from 2008 to 2022, the main trend is leveling from below, by reducing the gap from the national average in 2/3 of the middle and less developed regions (Zubarevich and Safronov, 2023). At the same time, the leveling effect is more noticeable in the Far East, the North Caucasus, and in most regions of the Center. However, no obvious connection between transfers and interregional economic inequality in GRP, measured by the Gini coefficient, is found, with the exception of a noticeable equalizing effect during the 2009 crisis and more clearly in the COVID year of 2020, when the volume of transfers to regions increased by more than one and a half times (Fig. 4).

(4) A comparative analysis of interregional inequality according to various indicators in the two largest post-Soviet countries by territory—Russia and Kazakhstan—shows that the trends in inequality in them had both similarities and differences (Figs. 5a, 5b). The dynamics for the 1990s and 2000s has already been discussed in a previous article by the authors (Zubarevich and Safronov, 2011). Therefore, we concentrate on analyzing changes for the 2010s and early 2020s. As has already been shown (see Fig. 2), interregional *economic inequality (according to GRP)* in Russia in the 2010s stabilized, and in the early 2020s began to grow again. In Kazakhstan since the late 2010s, there has been a decrease in interregional inequality in GRP. This is due to the intensification of the state's redistribution policy, and not to fluctuations in world prices for oil and gas resources, the impact of which is the same for the two countries. In 2018, Kazakhstan adopted the Strategic Development Plan of the Republic of Kazakhstan until 2025, where the priority of regional policy was to ensure managed urbanization.⁴ New directions of regional policy provided for the development of a wider range of growth centers: 4 urban agglomerations and 14 large urbanized zones, thereby stimulating the development of all regional

⁴ Plan for Strategic Development of the Republic of Kazakhstan until 2025, official website of the President of the Republic of Kazakhstan. https://www.akorda.kz/ru/legal_acts/decrees/ob-utverzhdanii-strategicheskogo-plana-razvitiya-respubliki-kazakhstan-do-2025-goda-i-priznanii-utrativshimi-silu-nekotoryh-ukazov-prezidenta (accessed January 25, 2024).

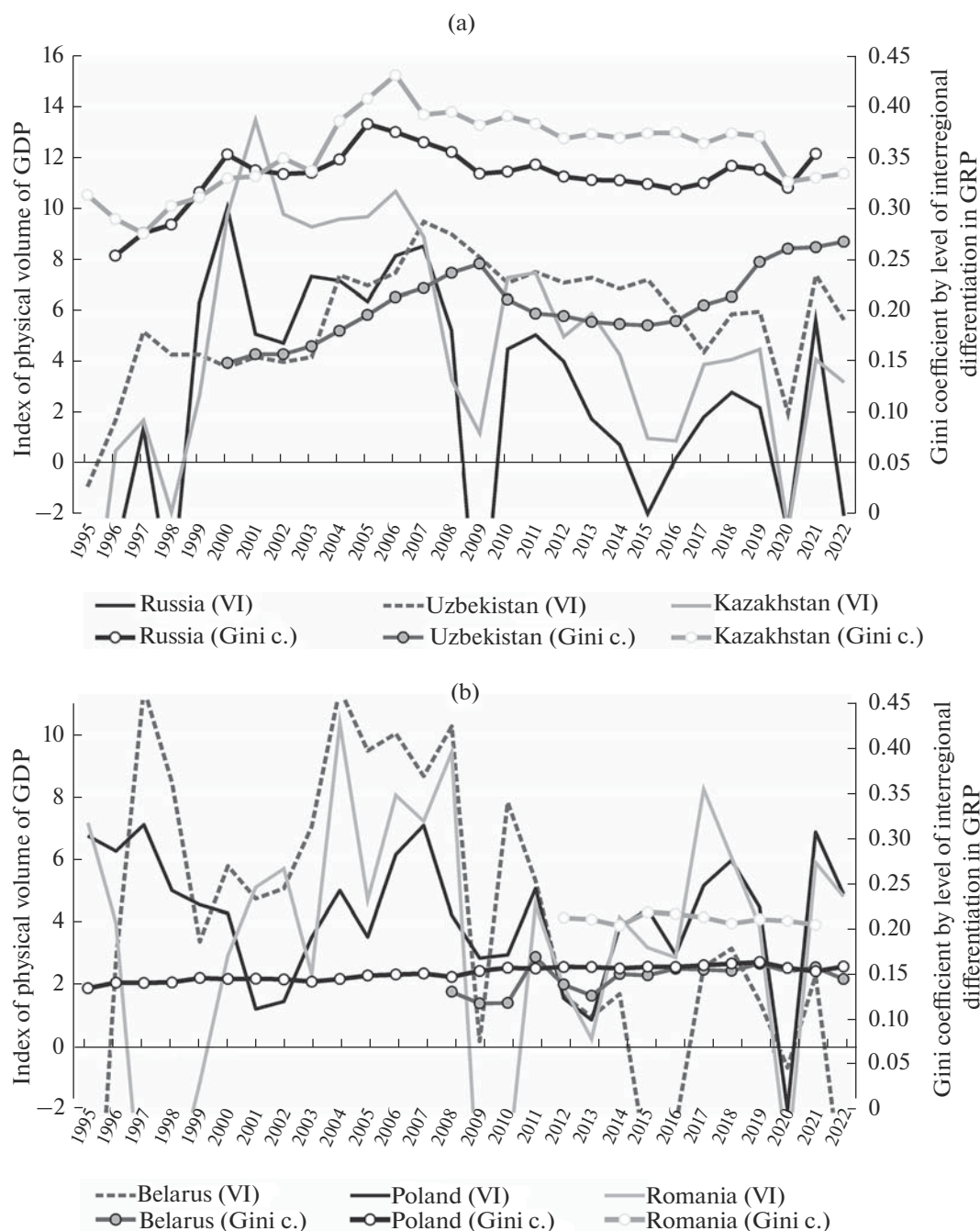


Fig. 3. Index of physical volume of GDP (VI) and Gini coefficient by level of interregional differentiation in GRP (Gini c.) for countries (a) with resource rent, (b) without resource rent.

Source: authors' calculations based on data from Rosstat, Bureau of National Statistics of Republic of Kazakhstan, National Statistical Committee of Republic of Belarus, Statistics Agency of Republic of Uzbekistan, World Bank.

centers, which helped mitigate interregional economic inequality. As a result, the two countries have become equal in terms of inequality, although previously it was higher in Kazakhstan.

The differences in the dynamics of interregional inequality in *investments* are even more noticeable: in Russia it has been growing since the mid-2010s, and in Kazakhstan it has been declining quite rapidly since

the late 2010s. For Russia, this can be explained by the growing share of Moscow in the total volume of investments in the country (from 15% in 2019 to 21% in 2021–2022). In Kazakhstan, interregional differentiation of investments is now lower than in Russia, although in the 2000s and 2010s, it was the other way around. The reasons for the leveling may be associated both with a decrease in the share of investments in oil-

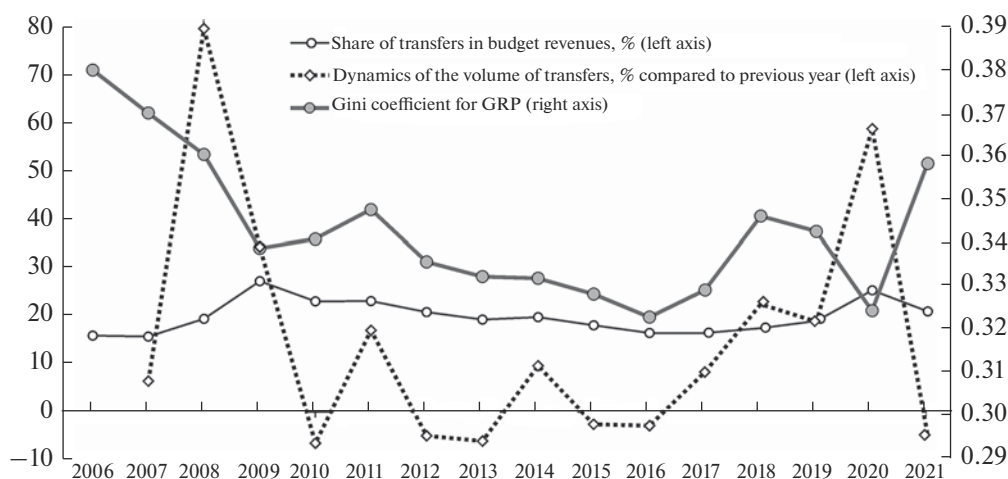


Fig. 4. Gini coefficient for interregional economic inequality (GRP), dynamics of transfers to regions (in % of previous year) and their share in revenues of consolidated budgets of regions of Russian Federation (%).

Source: authors' calculations based on data from Federal Treasury of Russian Federation and Rosstat.

producing regions and in the two capitals of Kazakhstan, and with an increase in investments in less developed regions at the expense of budgetary funds, but the very fact of a significant mitigation of interregional investment inequality is obvious.

Interregional differences in *housing construction* in Russia since the 2010s were lower than other economic indicators and changed less, despite fluctuations in individual years. Since the 2010s, 30–35% of all housing construction is concentrated in the metropolitan urban agglomerations of Moscow and St. Petersburg and Krasnodar krai. In Kazakhstan, interregional inequality in housing construction fell more significantly and noticeably only in the 2020s, almost reaching parity with Russia. Most likely, they began to build more housing in the regional centers of Kazakhstan, and not just in Astana and Almaty.

Interregional inequality in *retail trade* (data for organizations) in Russia and Kazakhstan was initially close, but then it decreased much faster in Russia, and now it is much less than in Kazakhstan. This is explained by the fact that in Russia there was a faster expansion of retail chains into the regions, especially into large cities, owing to which the whitewashing of retail took place. There are fewer large cities in Kazakhstan, and trade formats are still less modernized, so underestimation of the turnover of individual entrepreneurs and shadow retail (mainly open markets), largely preserved in the periphery, leads to more significant interregional differences.

Inequality in social indicators in the two countries also differs in level and dynamics. Interregional inequality by *money incomes per capita* was initially more significant in Russia, but it decreased significantly by the mid-2010s, owing to redistribution policies (increasing wages of public sector employees, etc.). However, since the second half of the 2010s, inequality grew slowly as the effect of the increases

wore off. At the same time, the gap between the incomes of the population of Moscow and the oil and gas producing autonomous districts of Tyumen oblast with respect to most regions of the country remains very strong. In Kazakhstan, income inequality was initially lower and its mitigation has been noticeable since the late 2000s. It is also necessary to take into account that regional income statistics are imperfect in all post-Soviet countries.

Interregional inequality in *wages* less income inequality and comparable for both countries. However, in Russia it has increased barely noticeably since the mid-2010s, and in Kazakhstan it decreased slightly in the early 2020s. In Russia, some increase in this inequality in recent years may be associated with a labor shortage due to demographic factors and faster growth of wages in more developed regions. The slight easing in interregional wage inequality in Kazakhstan is unlikely to be due to demographic reasons due to the younger age structure.

Interregional inequality by *poverty level* in Russia it is not as large, but it has grown since the mid-2010s and approached income inequality. At the same time, since 2018, Russia has been increasing targeted support for low-income families with children. Its increasing scale has contributed to a reduction in poverty levels in Russia as a whole since 2020, but is not sufficient to mitigate interregional inequality. In Kazakhstan, interregional poverty inequality grew until almost the end of the 2010s, and exceeded the indicators of Russia. Only since 2018 has it begun to decline rapidly, but is still higher than Russia's due to the higher share of the rural population and large families with low incomes, which are concentrated in less developed regions.

The Gini coefficient for *unemployment level*, measured using the methodology of the International Labor Organization (ILO), shows inequalities in

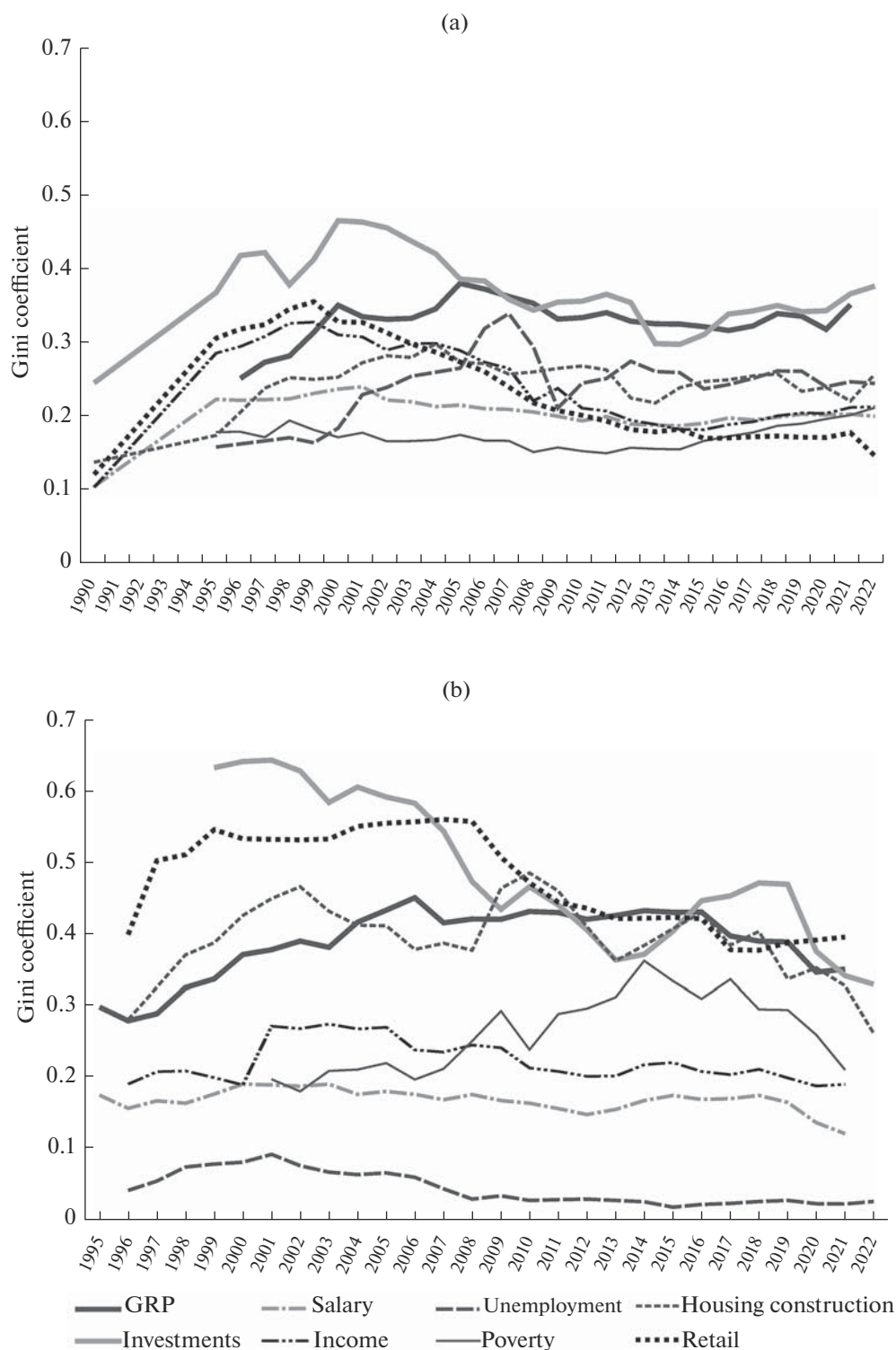


Fig. 5. Gini coefficient for interregional inequality for various socioeconomic indicators in Russia (a) and Kazakhstan (b).
Source: authors' calculations based on data from Rosstat and Bureau of National Statistics of Republic of Kazakhstan.

regional labor markets. However, comparison of differences in Russia and Kazakhstan is complicated by the influence of institutional factors. In Russia,

inequality increased significantly during the period of economic growth in the 2000s, then fell sharply during the 2009 crisis due to rising unemployment rates in

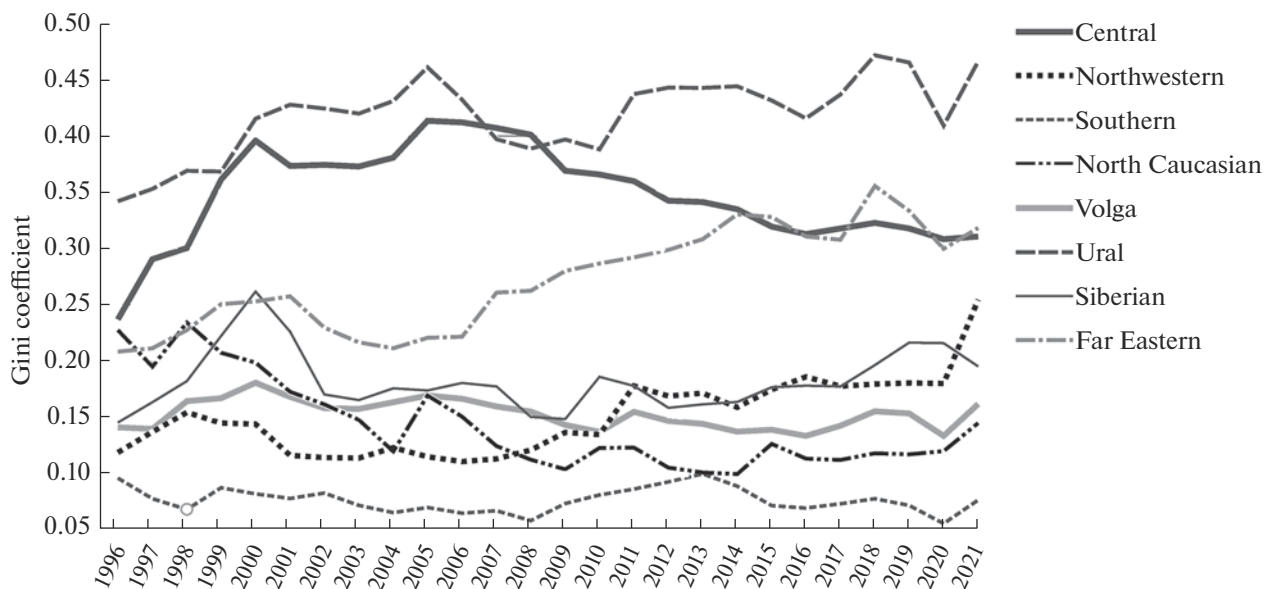


Fig. 6. Gini coefficient for interregional economic inequality (GRP) within federal districts of Russian Federation (within 2010 borders).

Source: authors' calculations based on Rosstat data.

more developed regions, and began to rise again as we emerged from the crisis. However, since 2013, changes in the Gini coefficient have been minimal, since in subsequent crises the unemployment rate in the regions remained almost unchanged; the main way labor markets adapted was not unemployment, but underemployment, widely used by employers during periods of crisis. In Kazakhstan, the role of institutional factors is even stronger: interregional differences in unemployment remain ultralow and are even declining, since all residents with livestock and land are legally considered self-employed and not unemployed. Due to the strong influence of institutional factors, it makes no sense to compare interregional inequality in two countries.

Thus, in recent years, in the two largest CIS countries, trends in interregional inequality have begun to differ: in Russia, according to most indicators, they are stable or growing, while in Kazakhstan they are declining. The reasons are clearer for Russia, where the volume of transfers since the 2010s has been insufficient to curb the growth of interregional differentiation in income and investment. In the case of Kazakhstan, further study of its regional policies, including changes in the scale of support for less developed regions, as well as features of investment policy and the budgetary system, is necessary to explain the reasons for the softening of interregional differences in recent years.

(5) Interregional inequalities can be assessed not only for all regions of the Russian Federation. An example is the study by E. Kolomak (2014), which showed that in the western part of Russia interregional economic and demographic differences grew faster

than in the eastern part. Below we present calculations of interregional economic inequality *within federal districts* and *economic areas* (EA). At the level of macroregions, the influence of the institutional factor, as well as the characteristics of their slicing, is even more highlighted (Fig. 6). The highest level of economic inequality is expected to be found in the Ural Federal District with a small number of regions, among them the Tyumen region, which receives oil and gas rent, and its autonomous okrugs (since 2011, they were counted separately, which led to an increase in inequality). Economic inequality of the regions of the Central Federal District until the end of the 2000s was comparable to the Ural ones due to the separation of Moscow; the institutional factor of the capital's rent plays an important role. Inequality only began to decline in the late 2010s owing to the growth of the economy of Moscow oblast and a number of other regions (Kaluga and Belgorod oblasts, etc.).

The economic inequality of regions of the Far Eastern Federal District has grown to the level of the Central Federal District due to the rapid development of Sakhalin oblast, where oil and gas production has sharply increased due to investment projects by global companies under production sharing agreements, and, partly, the Republic of Sakha (Yakutia), while regions of Zabaykalsky krai and the Jewish Autonomous Oblast remain depressed. The inequality of the regions of the Northwestern Federal District has increased in recent years due to the growing gap of St. Petersburg, which received additional benefits due to relocation of the headquarters of Gazprom and Gazpromneft. The oil-producing Nenets Autonomous Okrug (since 2011, it was considered separate from

Arkhangelsk oblast) also contributed to the growth of inequality. In terms of regional economic inequality, the Siberian Federal District is comparable to the Northwestern Federal District. The Volga, North Caucasian (where it even decreased owing to increased federal assistance in the form of transfers), and especially the Southern federal districts are characterized by minimal and more stable economic inequality in the regions. They do not have subjects that are strong growth drivers that increase inequality, although in the Volga Federal District, the Republic of Tatarstan could lay claim to such a role.

Regional inequality within *economic areas* largely coincides with differentiations by federal districts. The West Siberian EA, which includes the oil and gas producing Tyumen oblast with autonomous okrugs, has maximum and stable economic differentiation. Sharp interregional differences in the Central EA, as well as in the Central Federal District, are gradually decreasing. The same trend of increasing interregional differences between the regions of the Far Eastern EA, then its attenuation in the 2020s. Another similarity is the growing economic inequality of Northwestern EA regions. Growing economic disparities in the Northern EA in the early 2010s can be explained by the accelerated development of the oil-producing Nenets Autonomous Okrug, since there are no obvious growth drivers among other regions of the North. In other economic areas (Volga, North Caucasus, Central Chernozem, and Volga–Vyatka) natural conditions are more homogeneous and development is relatively high, so differentiation is small and quite consistent with the values of European countries.

CONCLUSIONS

Despite the growing number of studies of interregional inequality in Russia, a general idea regarding its scale and dynamics has not yet emerged. The level of interregional inequality in the European Union countries is significantly lower than in the most extensive post-Soviet countries. In the less developed Balkan countries, Turkey and Hungary it is slightly higher than in the developed EU countries. This confirms the conclusions of A.I. Treivish on greater interregional inequality in less developed countries. However, interregional inequality in the EU has been either stable or slightly declining since the early 2010s.

The main substantive conclusion is that the level of interregional inequality varies over time; there is no stable trend, since inequality, especially in post-Soviet countries, is influenced by many economic and institutional (including political) factors. A comparison of the level and dynamics of interregional inequality in large post-Soviet countries revealed different trends in the 2010s and early 2020s: in Russia since the late 2010s it grew, just like in Uzbekistan; it decreased in Kazakhstan and was small and stable in Belarus.

Assessments of the impact of macroeconomic dynamics (economic growth or decline) on interre-

gional inequality did not reveal an obvious relationship, which may be explained by the influence of other factors (regional economic structure, etc.). The impact of the state's redistribution policy, which can only be assessed for Russia, was more noticeable during the crises of 2009 and 2020, mitigating interregional inequality in GRP and personal income.

The dynamics of interregional inequality in Russia and Kazakhstan according to various economic and social indicators vary greatly and do not have a common trend: leveling off or increasing regional differences. Depending on the indicator chosen for changes in inequality, different results are obtained.

Measuring interregional inequality in Russia within federal districts and economic areas did not produce entirely obvious results. Maximum inequality is typical of the Ural Federal District due to the extremely high indicators of Tyumen oblast. The high level of inequality in the Central Federal District, due to Moscow's influence, has been decreasing. In the Far Eastern Federal District, inequality has increased and approached that in the Central Federal District. In recent years, inequality in the regions of the Northwestern Federal District has increased due to the influence of St. Petersburg. The remaining federal districts have a significantly lower scale. Interregional inequality within economic areas largely coincides with federal districts, except for rising inequality in the Northern EA.

In general, the analysis showed the inadequacy of simple judgments: interregional inequality is growing or decreasing. In Russia, it is really large, but the factors influencing it, the trends in the dynamics, and the scale of inequality according to individual indicators are different. This is important to understand not only within academic research, but also when developing and implementing regional policy.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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